Lie Groups — Notes

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We will follow the book "Lectures on Lie groups" by J. Frank Adams through chapter 5.56.

1.1 Introduction

Suppose that V, W are finite dimension vector spaces over $\mathbb{R}$. Notice that in such spaces all norms are equivalent. a function $f : U \to W$ where $U \subseteq V$ is differentiable if for all $x \in U$ there exists a linear map $df\ |_x = f'(x) : V \to W$ such that,

$$f(x+h) = f(x) + (df\ |_x)(h) + o(|h|)$$

Therefore $f' : U \to \hom(V, W)$ defined by $x \mapsto f'(x)$. We continue to define in such manner the next differentials recursively, $f'' : U \to \hom(V, \hom(V, W))$ etc. A function f is C^∞ (smooth) if the derivatives of any order exists and are continuous.

Definition 1.1 (Chart)**.** Let X be a topological space. **Chart** on X modeled on V, $\varphi_\alpha : X_\alpha \to \Omega_\alpha$ is a homeomorphism such that $X_\alpha \subseteq X$ is open and $\Omega_\alpha \subseteq V$ is open as well.

In Adams book chart is defined the other way around ($\Omega_\alpha \to X_\alpha$). Compatible charts $\varphi_\alpha, \varphi_\beta$ are charts such that there is a homeomorphism between them and are invertible.

An atlas on X is $\{\varphi_\alpha : X_\alpha \to \Omega_\alpha\}$ a set of charts, such that $\forall \alpha, \beta,\ \varphi_\beta \circ \varphi_\alpha^{-1}, \varphi_\alpha \circ \varphi_\beta^{-1}$ are smooth and $\bigcup_\alpha X_\alpha = X$. Two atlases are equivalent if their union is atlas.

Exercise 1.1. This is an equivalent relation.

Definition 1.2 (Smooth manifold)**.** A (smooth) manifold is a Hausdorff 2nd countable topological space X equipped with an equivalence class of atlases.

Given manifolds X, Y modeled on vector-spaces V, W, $X \times Y$ is a manifold modeled on $V \times W$. $\{\varphi_\alpha : X_\alpha \to \Omega_\alpha\}$, $\{\psi_\beta : Y_\beta \to \Omega'_\beta\}$ are atlases, then $\{\varphi_\alpha \times \psi_\beta : X_\alpha \times Y_\beta \to \Omega_\alpha \times \Omega_{\beta'}\}$ is an atlas in $X \times Y$. Equivalent atlases on X & on Y yield equivalent atlases on $X \times Y$. There exists a unique manifold structure on $X \times Y$ such that $X \times Y \to X$ & $X \times Y \to Y$ are smooth and for any Z manifold and $Z \to X \times Y$ is smooth iff the compositions are smooth.

Definition 1.3 (smooth map between manifolds)**.** Given X, Y are manifolds modeled on V, W and atlases $\{\varphi_\alpha\}$ and $\{\psi_\beta\}$. We say that $f : X \to Y$ is smooth if $\forall \alpha, \beta,\ \varphi_\beta \circ f \circ \psi_\alpha^{-1}$ is smooth.

Definition 1.4 (Lie group)**.** A Lie group G is a set equipped with:

1. a manifold structure
2. a group structure: a group operation $\mu : G \times G \to G$ defined as $\mu(a, b) = a \cdot b$ and inversion $i : G \to G$ defined as $i(x) = x^{-1}$.

such that μ and i are smooth.

Definition 1.5 (Homomorphism of Lie groups)**.** Let G, H be Lie groups, a function $f : G \to H$ that is smooth and a group Homomorphism is called Lie group Homomorphism.

Example 1.1. $(\mathbb{R}^n, +)$ is a Lie group.

Example 1.2. Define $(\mathbb{R}^n/\mathbb{Z}^n, +)$. In this case we can define $\forall x \in \mathbb{R}^n$ and $0 < \varepsilon < \frac{1}{2}$ we can define $\Omega_x = B(x, \varepsilon) \to X_x$ where X_x is its image in $\mathbb{R}^n/\mathbb{Z}^n$.

Example 1.3. Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$ denoted as Λ, and let $\mathrm{End}(V) = \hom_\Lambda(V, V)$ ($\mathrm{End}(V) \cong \mathbb{R}^{n \times n}$), and $\mathrm{Aut}(V) \subseteq \mathrm{End}(V)$ be the subset of invertible Λ-linear maps. $\mathrm{Aut}(V)$ is an open subset of $\mathrm{End}(V)$. In coordinates: $a, b \mapsto a \cdot b$ as matrix multiplication is a polynomial, hence smooth. It follows from Kramer's rule that $a \mapsto a^{-1}$ the matrix inversion is achieved by $a^{-1} = \frac{1}{\det a} \mathrm{adj}(a)$, and so it's smooth as well.

Remark. For X, Y are manifolds and $f : X \to Y$ is smooth if and only if any n and $S \subseteq \mathbb{R}^n$ open & smooth map $S \to X$, the composition $S \to X \to Y$ is smooth, if and only if for all $g : Y \to \mathbb{R}$ smooth, $f \circ g$ is smooth.

Definition 1.6 (Tangent space)**.** Tangent space to a manifold X at $x \in X$ is defined as,

$$T_x X = \{\gamma : (-\delta, \delta) \to X \text{ smooth}, \gamma(0) = x\}/\sim$$

where $\gamma_1 \sim \gamma_2 \iff \forall f : U \to \mathbb{R}$ where $x \in U$ is open neighborhood, smooth map, then $\frac{d}{dt} \mid_{t=0} f \circ \gamma_1 = \frac{d}{dt} \mid_{t=0} f \circ \gamma_2$.

Example 1.4. For $x \in X \subseteq V$ open, namely an open subset of a vector space, $T_x X = V$ via $\gamma(t) = x + tv$ for $v \in V$.

Smooth map of manifolds $f : X \to Y$ induces $f' = f_* = df \mid_x : T_x X \to T_{f(x)} Y$.

Given a manifold X & atlas $\{\varphi_\alpha : X_\alpha \to \Omega_\alpha\}$,

$$(\varphi_\alpha)^* : T_x X_\alpha \to T_{\varphi_\alpha(x)} \Omega_\alpha = V$$

Given a well defined vector-spaces structure on $T_x X$.

Definition 1.7 (Lie algebra)**.** Suppose that G is a Lie group and $e \in G$ its unit, we define $L(G) = T_e G$ (as a vector-spaces), Lie algebra of G. Also notated as $\mathrm{Lie}(G)$ or $\mathfrak{G}$.

Let G be a Lie group, $x \in G$, and define left multiplication,

$$L_x : G \to xg$$

defined a diffeomorphism $G \to G$ by $e \mapsto x$ hence a linear isomorphism $T_e G \xrightarrow{\sim} T_x G$, then left trivial injection $TG \cong G \times L(G)$.

definitions & theorems

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